

Sungmin Kang

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Education

Texas A&M University

Ph.D in Computer Science (Advisor: *Prof. Zhengzhong Tu*)

College Station, TX, United States

Aug. 2026 - Present

University of Southern California (USC)

M.S. in Electrical Engineering (Advisor: *Prof. Salman Avestimehr*)

GPA: 4.00/4.00

Los Angeles, CA, United States

Aug. 2024 - May. 2026

MS Honors Fellow

Sogang University

B.S. in Electronic Engineering

GPA: 3.91/4.3

Seoul, Republic of Korea

Mar. 2018 - Feb. 2024

Magna Cum Laude

Research Interests

My research interests lie in **AI safety** and **agentic AI systems**, with a focus on web agent safety, multi-agent information flow, and agent memory. Previously, I worked on LLM uncertainty quantification (UQ), and federated learning (FL).

Publications

- P1. **ReSCENE: Server-Side Replay for Structural Mitigation of Catastrophic Forgetting in Federated Continual Learning**
Sungmin Kang, Zhengzhong Tu[†], Sunwoo Lee[†]
Preprint, 2026
- C5. **Uncertainty as Feature Gaps: Epistemic Uncertainty Quantification of LLMs in Contextual Question-Answering**
Yavuz Faruk Bakman, Sungmin Kang, Zhiqi Huang, Duygu Nur Yaldiz, Catarina G Belém, Chenyang Zhu, Anoop Kumar, Alf Samuel, Daben Liu, Salman Avestimehr, Sai Praneeth Karimireddy
The Fourteenth International Conference on Learning Representations (ICLR 2026)
- C4. **GEM: A Scale- and Distribution-Aware Sparse Fine-Tuning Framework for Effective Downstream Adaptation**
Sungmin Kang, Jisoo Kim, Salman Avestimehr, Sunwoo Lee
The 40th Annual AAAI Conference on Artificial Intelligence (AAAI 2026)
- C3. **Layer-wise Update Aggregation with Recycling for Communication-Efficient Federated Learning**
Jisoo Kim, Sungmin Kang, Sunwoo Lee
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)
- C2. **TruthTorchLM: A Comprehensive Library for Predicting Truthfulness in LLM Outputs**
Duygu Nur Yaldiz*, Yavuz Faruk Bakman*, Sungmin Kang, Alperen Ozis, Hayrettin Eren Yildiz, Mitash Shah, Zhiqi Huang, Anoop Kumar, Alf Samuel, Daben Liu, Sai Praneeth Karimireddy, Salman Avestimehr
The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025) [Github]
- C1. **Reconsidering LLM Uncertainty Estimation Methods in the Wild**
Yavuz Faruk Bakman*, Duygu Nur Yaldiz*, Sungmin Kang, Tuo Zhang, Baturalp Buyukates, Salman Avestimehr, Sai Praneeth Karimireddy
63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)
- M1. **Uncertainty Quantification for Hallucination Detection in Large Language Models: Foundations, Methodology, and Future Directions**
Sungmin Kang, Yavuz Faruk Bakman, Duygu Nur Yaldiz, Baturalp Buyukates, Salman Avestimehr
IEEE BITS the Information Theory Magazine, 2025

Work Experience

Research Intern, Argonne National Laboratory (Host: *Dr. Kibaek Kim*)

Jun. 2026 - Aug. 2026

- **Agentic Framework-Native Code Generation:** Designed an LLM-agent pipeline that reads an FL research paper and implements its algorithm natively within existing FL frameworks (APPFL, Flower, NVFlare)
- **Realistic FL Simulator:** Built a simulator that faithfully reproduces FL environments, supporting settings from large-scale cross-device to high-performance computing and reporting system cost (compute, communication, queueing)

Research Experience

Research Assistant, Texas A&M University (Advisor: Prof. Zhengzhong Tu) Aug. 2026 - Present

- **Web Agent Safety:** Analyzing how benign-looking images under an attacker’s control can steer the decisions of web agents, and exploring zero-shot defenses against such attacks
- **Self-Improving Safety Guards:** Exploring safety guards for agents that improve themselves through recursive self improvement (RSI) to defend against evolving attacks
- **Forgetting in Agentic Memory:** Exploring when long-horizon agents should remove stale context rather than compress it, based on its relevance to the global goal of a hierarchical task

Research Assistant, University of Southern California (Advisor: Prof. Salman Avestimehr) Oct. 2024 - May. 2026

- **Representation-Based UQ [C5]:** Defined epistemic uncertainty of LLMs as semantic feature gaps in hidden representations relative to an idealized model, and measured these gaps in contextual QA using contrastive prompts
- **UQ Evaluation [C1]:** Evaluated existing UQ methods under realistic deployment conditions: threshold calibration under distribution shift, robustness to input variation, long-form responses, and ensembling
- **Open-Source UQ Library [C2]:** Built TruthTorchLM, a library implementing 25+ UQ methods with unified support for calibration, evaluation, long-form responses, and contextual QA
- **UQ Survey [M1]:** Authored a survey that categorizes 35 UQ methods and outlines open problems and future directions

Research Intern, Inha University (with Prof. Sunwoo Lee) May. 2024 - Dec. 2025

- **Federated Continual Learning [P1]:** Proposed a method that shifts remembering from clients to the server, which replays condensed surrogates of past tasks, with temporal herding choosing which to keep in a fixed-size buffer
- **Communication-Efficient FL [C3]:** Designed a method that uploads layers with large updates relative to their weights and reuses previous updates for layers with small ones
- **Parameter-Efficient Fine-Tuning [C4]:** Proposed a sparse fine-tuning framework that uses the gradient-to-weight ratio to identify and tune only the important parameters for effective downstream adaptation

Honors and Awards

MS Honors Fellowship, USC	May 2026
MS Outstanding Academic Achievement Award (one master’s student department-wide), USC	May 2026
Student Travel Scholarship & Volunteer, AAAI-26	Jan. 2026
Best Poster Award (out of 110+ teams), ECE 15th Annual Research Festival, USC	Oct. 2025
Best Poster Award (out of 50+ teams), Viterbi PhD Visit Day, USC	Apr. 2025
3 rd Prize (out of 50+ teams), Senior Thesis Project Contest, Sogang University	Dec. 2023
3 rd Prize (out of 40+ teams), Capstone Design Project Contest, Sogang University	Jun. 2023
Daesang Foundation Scholarship (Merit-based scholarship, 5M KRW/semester)	Spring 2019 - Fall 2023

Teaching Experience

Grader, EE 503: Probability for Engineers by Prof. Kosko Bart, USC	Fall 2025
Teaching Assistant, EEE 4171: AI Communications by Prof. Hongseok Kim, Sogang University	Spring 2024
Teaching Assistant, COR 1010: AI Programming by Prof. Naeun Jang, Sogang University	Summer 2023

Academic Services

Reviewer: ICLR 2027, IEEE Transactions on Artificial Intelligence

Invited Talk: “Federated Learning Implementation through Socket Communication”, Sogang University May 2024

Leadership Experience

Sergeant, Battalion Leader, Korean Augmentation to the US Army (KATUSA), Camp Hovey Oct. 2019 - May 2021